

```
command_pnDTHD_pci_b_command-start
137 mysqlq23509      mysqlq __1cQclose_connection6FpnDTHD_lb_v_
connection-done
138 mysqlq23509      mysqlq      handle_one_connection connection-
start
139 mysqlq23509      mysqlq __1cMmysql_delete6FpnDTHD_pnKTABLE_
LIST_pnItem_pnList_sqi_list_LXb_b_ delete-done
140 mysqlq23509      mysqlq __1cHhandlerNha_delete_row6MpkC_i_
delete-row-done
```

As expected, the function names are mangled. You can view the demangled names with `c++filt`:

```
pfexec dtrace -l -m:mysq! | c++filt
```

...which will show the demangled functions:

ID	PROVIDER	MODULE	FUNCTION NAME
56209	mysqlq706	mysqlq	bool dispatch_command(enum_server_com
mand.THD*.char*, unsigned) command-done			
56210	mysqlq706	mysqlq	bool dispatch_command(enum_server_com
mand.THD*.char*, unsigned) command-start			
56211	mysqlq706	mysqlq	void close_connection(THD*, unsigned, bool)
connection-done			
56212	mysqlq706	mysqlq	handle_one_connection connection-
start			
56213	mysqlq706	mysqlq	bool mysql_delete(THD*, TABLE_,
LIST*.Item*.st_sqi_list*.unsigned long, unsigned long long, bool) delete-done			
56214	mysqlq706	mysqlq	int handler:ha_delete_row(const unsigned
char*) delete-row-done			
56215	mysqlq706	mysqlq	int handler:ha_delete_row(const unsigned
char*) delete-row-start			
56216	mysqlq706	mysqlq	int mysql_execute_command(THD*) delete-
start			
56217	mysqlq706	mysqlq	unsigned long filesort(THD*.TABLE*.st_
sort_field*.unsigned.SQL_SELECT*.unsigned long, bool, unsigned long*)			
filesort-done			
56218	mysqlq706	mysqlq	unsigned long filesort(THD*.TABLE*.st_
sort_field*.unsigned.SQL_SELECT*.unsigned long, bool, unsigned long*)			
filesort-start			
56219	mysqlq706	mysqlq	int handler:ha_external_lock(THD*.int)
handler-rdlock-done			
56220	mysqlq706	mysqlq	int handler:ha_external_lock(THD*.int)
handler-rdlock-start			
56221	mysqlq706	mysqlq	int handler:ha_external_lock(THD*.int)
handler-unlock-done			
56222	mysqlq706	mysqlq	int handler:ha_external_lock(THD*.int)
handler-unlock-start			
56223	mysqlq706	mysqlq	int handler:ha_external_lock(THD*.int)
handler-wrlock-done			
56224	mysqlq706	mysqlq	int handler:ha_external_lock(THD*.int)
handler-wrlock-start			
56225	mysqlq706	mysqlq	bool mysql_insert(THD*.TABLE_
LIST*.List*.List>*.List&.enum_duplicates, bool) insert-done			
56226	mysqlq706	mysqlq	int handler:ha_write_row(unsigned char*)

```
insert-row-done
56227 mysqlq706      mysqlq int handler:ha_write_row(unsigned char*)
insert-row-start
56228 mysqlq706      mysqlq      void select_insert:abort() insert-select-
done
56229 mysqlq706      mysqlq bool select_insert:send_eof() insert-select-
done
56230 mysqlq706      mysqlq int mysql_execute_command(THD*) insert-
select-start
56231 mysqlq706      mysqlq int mysql_execute_command(THD*) insert-
start
56232 mysqlq706      mysqlq bool multi_delete:send_data(List&) multi-
delete-done
56233 mysqlq706      mysqlq int mysql_execute_command(THD*) multi-
delete-start
56234 mysqlq706      mysqlq bool multi_update:send_data(List&) multi-
update-done
```

A better query logger

Using one of the static probes, we shall now write a better query logger, containing information such as username and connection ID.

```
#/usr/sbin/dtrace -qws
```

```
#pragma D Option strsize=1024
```

```
dtrace: BEGIN
```

```
{
    printf("Logging to file. Hit Ctrl-C to end.\n");
    freopen("/tmp/mysql.log");

}

mysqlq:::query-start /* using the mysql provider */
{
    self->query = copyinstr(arg0); /* Get the query */
    self->connid = arg1; /* Get the connection ID */
    self->db = copyinstr(arg2); /* Get the DB name */
    self->who = strjoin(copyinstr(arg5), strjoin("@", copyinstr(arg4))); /* Get the
    username */
}
```

```
printf("%Y\t%20s\t Connection ID: %d\t\t Database: %s\t\t Query: %s\n",
    walltimestamp, self->who, self->connid, self->db, self->query);
```

```
}
```

This will store logs into a file, `/tmp/mysql.log`:

```
2009 Feb 23 15:42:38      root@localhost      Connection ID: 3      Database:
Query: select user()
2009 Feb 23 15:42:41      root@localhost      Connection ID: 3      Database:
Query: show engines
2009 Feb 23 15:42:51      root@localhost      Connection ID: 3      Database:
Query: create table fo_bawrr(isa INTEGER)
2009 Feb 23 15:42:55      root@localhost      Connection ID: 3      Database:
Query: SELECT DATABASE()
```